

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element J Initial Template

How to use this template

In order to earn a 5 on this element, you will need to collect project evaluation from 3 or more stakeholders and 3 or more demonstrably-qualified field experts. Project evaluation includes providing feedback on the prototype, reviewing the testing results, providing feedback on the analysis of the testing results, and providing recommendations for improvement. Project evaluation should address 2 or more aspects of your design or engineering design process. Duplicate the evaluator template as needed if you have more evaluators than are listed in the template below.

Element J: Documentation of Evaluation

Stakeholder Evaluation #1

- Mr. L
- Mr.L is a stakeholder because he teaches multiple STEM and engineering related classes in the room, and is the recipient of the 3d printer filament storage system. The smooth operation of his class is extremely important to him, as it allows him to teach with fewer interruptions so that the students can learn better.
- Documentation of the project evaluation, which could include copies of emails, transcripts of conversations, filled out feedback forms, etc.

Transcript:

Mr. L: I like the vertical storage, especially because we have something like that now for the filaments – that rack by the shelf.

Alex & Aidan: What about the horizontal storage? That's supposed to be the long-term storage.

Mr. L: How would you get the rolls from there? Is there enough space?

Name: _____

Date: _____

Class: _____

Alex & Aidan: There's more than enough space on top. The rolls would be organized however you like, and you would take one roll over the top of the posts.

Mr. L: You could also use PVC pipe, it's lighter and cheaper. Also, you could screw the pipe into the wood so that if you need a roll towards the bottom you can easily access it. Everything looks good though!

Stakeholder Evaluation #2

- Hayden Alexander (Student)
- He is Mr. L class and commonly uses the printers.
- Aidan: What do you think about the filament storage for Mr. L's class.

Hayden: I usually print projects that use a lot of filament or use different color filaments and changing the filament is a pain because I have to search through the stacks of filaments to find what I need which takes forever.

Aidan: Would this kind of system help? (explains our project idea)

Hayden: Yeah that would make things a lot easier. Having a storage system right above the printers and easy to see would make things a whole lot easier for me.

Aidan: Do you have any ideas on what to improve in the design?

Hayden: No it sounds like a good idea

Aidan: Alright thanks.

Stakeholder Evaluation #3

- Anonymous student
- He is Mr. L class and commonly uses the printers.
- Aidan: What do you think of our idea

Anonymous student: I think it's good and really interesting.

Aidan: any advice

Anonymous student: No I think that's it.

Name: _____

Date: _____

Class: _____

Field Expert Evaluation #1

- Name of evaluator: NOVA Fab Lab
- Richard Sewell is the Nova Fab Labs Fabrication Coordinator
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3D printer filament storage



C

Carter Stone <carterstone2005@gmail.com>
to novafablab ▾

Mar 21, 2024, 12:36 PM (13 days ago)



Dear NOVA Fab Labs,

We are writing this email to ask a couple questions about your 3D printer filament storage. We are students at Marshall high school, and I am working on a design project in my STEM engineering class. Our problem to solve is a 3D printer filament storage system for another STEM lab and teacher in the school. We were curious what your solution to this problem has been at the NOVA Fab Labs, and what methods you have utilized to negate this issue. Thank you so much for your time, and if you have any questions about our design, or qualifiers about the design problem, feel free to reach out.

Sincerely,

-Carter Stone

**Sewell, Richard A.** <rsewell@nvcc.edu>
to me, novafablab ▾

Mar 26, 2024, 5:30 PM (8 days ago)



Hey Carter,

Interesting project, and I'd be happy to lend my input.

We currently use a handful of different manufacturers for our spools.

- Most are Stratasys ABS and QSR support spools for our large F-Series printers. These are sealed on shipment, and we make it a point not to unseal them until absolutely necessary – once we have rolls unsealed and not being used in the machines - we keep them in vacuum sealed bags with desiccant packs and humidity meters.
- Our PLA is for our Prusa MK4's, and we are less finicky with this material – partly b/c we go through spools quickly, and also because those machines are very forgiving when it comes to printing slightly wet filament.
- The Markforge machines come with their own dry boxes, so the nylon onyx material stays very dry – in fact one spool has been in there for more than a year and is still good to go – of course these dry boxes contain silica beads as well which we can check for color changes.

Hope that helps – happy to answer additional questions should you need it.

- Richard

Richard Sewell

NOVA SySTEMic

Fabrication Lab Coordinator

Field Expert Evaluation #2

- Name of evaluator:
- Justification for how this evaluator is a demonstrably-qualified field expert (professionals who spend at least a portion of their time working in a field directly related to your project):
- Documentation of the project evaluation, which could include copies of emails, transcripts of conversations, filled out feedback forms, etc.

Name: _____

Date: _____

Class: _____

Synthesis of the Project External Evaluation

Summarize all of the evaluators' comments in a way that highlights meaningful trends with specifics mentioned. Answer the questions, "What does the external evaluation mean?" and "What are you taking away from the external evaluation?"

We had four sources for our external evaluation, Mr. L, Hayden Alexander – a student taking a robotics class in the same classroom, another anonymous student in the classroom, and NOVA Fab Labs. There were a couple common threads through all the evaluations. First, the evaluators liked the ease of organization in our design, and the proximity to the 3D printer for a convenient workflow. Second, though they liked the organizational ability, some sources were concerned about the actual difficulty of accessing those filaments from the shelves. For example, even though the filaments are neatly organized by color and easy to find, would it be difficult to get the red filaments or blue filaments off of the shelf? The evaluations from NOVA Fab Labs and Mr. L also made it clear that there was room for improvement when it comes to the materials being used for the build. Overall though, the evaluators liked the design and believed in its ability to solve the classroom filament issues. In terms of takeaways, clearly we should continue to prioritize organization into the design by using our dowel system for storing filaments. Additionally, we should keep the actual storage system very close to the 3D printers for the convenience of everyone in the classroom. To address the concerns about ease of access, we will switch to dowels/pipes that can be screwed into the shelf using 3D printed connections. Finally, we will switch from wooden dowels to cheaper and lighter PVC, which will work better for our design in this context. More broadly, we should definitely keep the general outline of our shelving system, just changing those smaller details while we continue to work on it.